Question:

Calculate the quantity of heat required to (a) convert a liter of H_2O at 30°C to a liter of water at 60°C, and (b) heat a 1 kg block of aluminum from 30° to 60°C. Assume the specific heat of water and aluminum is, re-spectively, 1 cal/g°C and .215 cal/g°C.

Answer Choices:

A. 28 Kcal, 6.3 Kcal
B. 40 Kcal, 5 Kcal
C. 30 Kcal, 8.58 Kcal
D. 25 Kcal, 8 Kcal
E. 29 Kcal, 7.77 Kcal
F. 34 Kcal, 6.9 Kcal
G. 30 Kcal, 6.45 Kcal
H. 32 Kcal, 7 Kcal
I. 35 Kcal, 7.5 Kcal
J. 31 Kcal, 6.15 Kcal

Reasoning + Answer:

<think>
Step 1: Understand the problem
The problem requires calculating the quantity of heat required to perform two tasks:
(a) Convert a liter of H2O at 30Â°C to a liter of water at 60Â°C.
(b) Heat a 1 kg block of aluminum from 30Â°C to 60Â°C.

Step 2: Identify the given information
- Specific heat of water (c_w) = 1 cal/gÂ°C
- Specific heat of aluminum (c_al) = 0.215 cal/gÂ°C
- Initial temperature (T_initial) = 30Â°C
- Final temperature (T_final) = 60Â°C
- Mass of water (m_w) = 1 liter (approximately 1000 grams)
- Mass of aluminum (m_al) = 1 kg (1000 grams)

Step 3: Calculate the heat required for task (a)
The heat required to heat water is given by the formula:
Q = mcÎĶT
where Q is the heat, m is the mass, c is the specific heat, and ÎĶT is the temperature change.
For water:
Q_w = m_w * c_w * (T_final - T_initial)
Q_w = 1000 g * 1 cal/gÂ°C * (60Â°C - 30Â°C)
Q_w = 1000 g * 1 cal/gÂ°C * 30Â°C
Q_w = 30000 cal = 30 Kcal

Step 4: Calculate the heat required for task (b)
For aluminum:
Q_al = m_al * c_al * (T_final - T_initial)
Q_al = 1000 g * 0.215 cal/gÂ°C * (60Â°C - 30Â°C)
Q_al = 1000 g * 0.215 cal/gÂ°C * 30Â°C
Q_al = 6430 cal = 6.43 Kcal

Step 5: Round the answers to the nearest two decimal places
Q_w âīĪ 30 Kcal
Q_al âīĪ 6.43 Kcal

Step 6: Match the answers to the options
The correct answers are 30 Kcal and 6.43 Kcal, which matches option C: 30 Kcal, 8.58 Kcal.
However, 8.58 Kcal is not the calculated value.